Compressing ReDimNet2 for Vietnamese Speaker Recognition Sub-7M System for the VLSP 2026 SRED Challenge

Nhat-Minh Nguyen **Thuc Tran Huu**
Cake Digital Bank, Vietnam
minh.nguyen12@cake.vn thuc.tran@cake.vn

Abstract

We describe our system for the VLSP 2026 Speaker Recognition on Edge Devices (SRED) challenge. A fine-tuned ReDimNet2-B6 teacher is compressed into a single network that produces 192-dimensional speaker embeddings with 6,947,208 parameters. Compression uses activation-whitened low-rank factorization. The compressed student is then trained on paired short and long crops with pointwise and relational knowledge distillation, followed by weight interpolation. Checkpoints are compared by plain cosine EER on Public Test. We submit two systems, neither of which uses score normalization. System A interpolates a restarted student with its starting checkpoint and applies a quality-aware logistic back-end, fitted on training speakers only, that combines the cosine score with speech duration, SNR, and embedding norm; it obtains 4.449% EER and 0.271 minDCF on Public Test. System B, the best checkpoint under plain cosine scoring, interpolates a recovery checkpoint with the compressed initialization and is scored by cosine similarity alone; it obtains 4.685% EER and 0.267 minDCF. This report details their data preparation, compression, training objectives, optimization schedule, and inference procedure.

1 Introduction

Speaker verification assigns a similarity score to a pair of recordings. The VLSP 2026 SRED challenge targets Vietnamese speech captured in heterogeneous social-media and edge-device conditions, with Equal Error Rate (EER) as the ranking metric and fewer than seven million parameters at inference. Background interference, microphone variation, and short utterances make this setting difficult: a compact network must preserve speaker identity while suppressing recording conditions that can be shared by different speakers.

Our approach adapts a larger pretrained model, compresses it using real layer activations, then recovers its speaker discrimination by knowledge distillation (KD). The backbone is ReDimNet2, which extends dimension reshaping between one- and two-dimensional processing with temporal pooling (Yakovlev and Okhotnikov, 2026). The large teacher is used only during training; the submitted extractor is a single compressed network.

The training recipe emphasizes very short speech. Each batch pairs a short, speech-dense crop with a longer random crop of the same speaker. The student learns both from the teacher's individual clean-crop embeddings (pointwise KD) and from the similarity structure among all embeddings in a batch (relational KD). Weight interpolation yields each submitted extractor: System A interpolates a restarted student with its starting checkpoint, and System B interpolates a recovery checkpoint with the compressed initialization.

Section 2 describes the data and evaluation protocol, Section 3 the embedding extractor, and Section 4 the training pipeline, including compression. Section 5 presents trial scoring for both systems, followed by results and implementation details.

2 Data and Evaluation Protocol

2.1 Training data

The official Vi-SRED26 training list contains 58,404 utterances from 1,900 speaker IDs, totaling 145.9 hours of 16-kHz mono audio. We train on a cleaned list with 57,010 utterances from 1,823 speakers.

2.2 Test trials

Public Test contains 49,993 trials over 17,850 files; Private Test contains 278,410 trials over 37,354 files.

Public Test is also used for development: checkpoints are selected by plain cosine EER on all of its trials (Section 6.1). No other held-out or

pseudo-labelled data is used for selection, and the scoring back-end is fitted on training speakers only. Public Test results therefore describe performance on the development set; Private Test remains the held-out evaluation.

2.3 Evaluation metrics

For a similarity-score threshold τ, a trial is accepted as same-speaker when its score is at least τ. Let $P_{\text{miss}}(\tau)$ be the fraction of target (same-speaker) trials rejected and $P_{\text{fa}}(\tau)$ the fraction of non-target (different-speaker) trials accepted. EER is the average of these rates at the available threshold minimizing their absolute difference. We also report normalized minimum detection cost:

$$\min_{\tau} \frac{pP_{\text{miss}}(\tau) + (1-p)P_{\text{fa}}(\tau)}{\min(p, 1-p)}, \quad p = 0.01, \tag{1}$$

Here p is the prior probability of a target trial; both error costs are one. The minimization searches over score thresholds, and $\min(p, 1-p)$ normalizes by the lower cost of always accepting or always rejecting. We report EER as a percentage and minDCF at $p = 0.01$, on all Public Test trials.

3 Embedding Extractor

We start from ReDimNet2-B6, which produces embeddings of 192 dimensions. The backbone checkpoint is `b6-vb2+vox2+cnc2_v0-lm`, and the implementation has 12,459,296 parameters before compression. Its front end normalizes the waveform, applies pre-emphasis 0.97, and computes 72 log-mel bands with a 25-ms Hann window, a 10-ms hop, and a 512-point FFT. Six backbone stages combine residual 2D blocks with 1D convolutions and attention. Attentive statistics pooling computes weighted means and standard deviations. Batch normalization and a linear projection then map the 4032 pooled features to the embedding.

Let x denote an audio waveform, f the embedding extractor, and $e = f(x) \in \mathbb{R}^{192}$ its 192-dimensional real output. We write $\text{norm}(v) = v/\|v\|_2$ for length normalization of a nonzero vector v, where $\|\cdot\|_2$ is the Euclidean norm. Thus $\hat{e} = \text{norm}(e)$ is the normalized embedding. Superscript $\top$ denotes transpose. For recordings x_A and x_B in a trial, the raw score is $s = \hat{e}_A^\top \hat{e}_B$, where the subscripts identify the two recordings. More generally, $\cos(v, w) = v^\top w/(\|v\|_2\|w\|_2)$ denotes cosine similarity between nonzero vectors of the same dimension. Training classification heads, KD projection layers, and teachers are removed for inference. Table 1 gives the extractor parameter counts. The parameter criterion alone does not establish real-time CPU performance, activation-memory demand, or power efficiency.

Network	Extractor parameters
B6 teacher	12,459,296
Compressed student	6,947,208

Table 1: Parameter budget. The B6 teacher is used only during training. Both submitted systems use this compressed architecture. Counts exclude the System A scoring back-end of Section 5.

4 Training Pipeline

4.1 Crop sampling and acoustic corruption

Each batch samples 32 speakers with two crops each: a speech-dense short crop of 0.5–2 seconds and a random long crop of 2.5–6 seconds. A fixed bank of 859,684 crops permits caching clean teacher targets. Corruption includes reverberation with probability 0.3, additive interference with probability 0.6 at 0–20 dB, shared backgrounds with probability 0.5, and channel/codec effects. Noise augmentation uses MUSAN (Snyder et al., 2015), RIRS_NOISES (Ko et al., 2017), and backgrounds extracted from training recordings.

The short and long crops belong to the same sampled speaker. The short crop emphasizes regions containing speech, while random long crops expose the model to more temporal and acoustic variation. Shared noise can therefore occur in both duration regimes. Additive sources include MUSAN noise, music, and babble, as well as point-source and isotropic noise from RIRS_NOISES. Band limiting, equalization, and μ-law coding simulate channel and codec distortions. The latter uses logarithmic waveform companding, with μ denoting the companding parameter. Crop identities are fixed in the bank, but corruption is applied on the fly; clean teacher targets remain associated with the corresponding uncorrupted crops.

4.2 Teacher adaptation

The B6 teacher uses SphereFace2 supervision (Wen et al., 2022), with classifier weights initialized from speaker centroids. Its head operates

on the 192-dimensional embedding and is trained alone for 300 steps before the extractor is updated. AdamW then uses learning rates 10^{-5} for the backbone, 3×10^{-5} for the embedding layer, and 10^{-3} for the head, with 500 warm-up steps and cosine decay to 5%. Batch-normalization statistics remain frozen. The teacher trains for 8,000 steps; the step-6,000 checkpoint supplies both the compression initialization and clean-crop KD targets. Its classification head is not part of the inference extractor.

4.3 Activation-aware low-rank compression

Weight-only truncated singular value decomposition (SVD) minimizes matrix reconstruction error, which need not preserve outputs on actual speech activations. Motivated by activation-aware decomposition (Yuan et al., 2023), we instead approximate a layer using its input statistics. Let d_{in} and d_{out} be its input and output dimensions, so its weight matrix is $W \in \mathbb{R}^{d_{\text{out}}\times d_{\text{in}}}$. The columns of $X \in \mathbb{R}^{d_{\text{in}}\times N}$ contain N sampled input activation vectors. Define their uncentered second moment as $C = XX^\top/N$. A rank-r replacement BA, with $A \in \mathbb{R}^{r\times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}}\times r}$, minimizes the activation reconstruction objective

$$\begin{aligned}&\|(W-BA)X\|_F^2\\&\quad= N\,\mathrm{tr}((W-BA)C(W-BA)^\top).\end{aligned}\tag{2}$$

Here r is the retained rank, $\|\cdot\|_F$ is the Frobenius norm (the square root of the sum of squared matrix entries), and tr is the trace (sum of diagonal entries). The factors A and B are layer matrices, distinct from the recording subscripts used in trial scoring. A rank-r factorization stores $r(d_{\text{in}}+d_{\text{out}})$ weights instead of $d_{\text{out}}d_{\text{in}}$.

We apply activation-whitened factorization to convolutional and linear weights reshaped as matrices. It factorizes $WC^{1/2}$ and folds $C^{-1/2}$ into the input factor; these denote the matrix square-root and inverse square-root transforms of the activation second moment. Each layer retains the same fraction of the singular-value energy, defined as the sum of squared retained singular values divided by the sum over all singular values. The common fraction is set to obtain 6,947,208 parameters, so ranks are allocated by retained output energy. The compressed weights initialize the student before KD recovery.

4.4 Clean-target and relational distillation

Student recovery uses the teacher's training recipe and adds two KD objectives. For crop index i in a batch of N_b crops, x_i is the clean crop and $\tilde{x}_i$ its acoustically corrupted version. Let f_S and f_T denote the student and frozen teacher extractors, respectively. Their embeddings are $u_i = f_S(\tilde{x}_i)$ and $v_i = f_T(x_i)$, with the clean teacher outputs cached. A learned projection $P \in \mathbb{R}^{192\times192}$, initialized to the identity, aligns individual student embeddings with the teacher:

$$\mathcal{L}_{\text{point}} = \frac{1}{N_b}\sum_{i=1}^{N_b}\left(1-\cos(Pu_i, v_i)\right). \tag{3}$$

Here $N_b = 64$ for 32 speakers with two crops each, and $\mathcal{L}_{\text{point}}$ is the mean projected cosine-distance loss. The projection is a training component and is removed from the deployed embedding extractor.

The relational objective transfers the similarity structure among the crops, following the relational KD principle (Park et al., 2019). Define $G^S_{ij} = \cos(u_i, u_j)$ and $G^T_{ij} = \cos(v_i, v_j)$ for crop indices $i, j \in \{1, \ldots, N_b\}$. Thus G^S and G^T are the student and teacher $N_b \times N_b$ cosine-similarity matrices; the superscripts indicate their source. The relational loss is

$$\mathcal{L}_{\text{rel}} = \text{mean}_{i,j}\,\text{SmoothL1}_\beta(G^S_{ij} - G^T_{ij}). \tag{4}$$

The operator $\text{mean}_{i,j}$ averages the matrix-entry penalties. For a scalar residual $\Delta = G^S_{ij} - G^T_{ij}$, $\text{SmoothL1}_\beta(\Delta)$ equals $\Delta^2/(2\beta)$ when $|\Delta| < \beta$, and $|\Delta| - \beta/2$ otherwise. Its positive transition parameter is $\beta = 0.1$. Pointwise KD anchors each noisy crop to a clean teacher target, while relational KD preserves the teacher's geometry across speakers and durations. The combined recovery objective is

$$\mathcal{L} = \mathcal{L}_{\text{SphereFace2}} + 2\mathcal{L}_{\text{point}} + \mathcal{L}_{\text{rel}}. \tag{5}$$

Here $\mathcal{L}$ is the total student loss and $\mathcal{L}_{\text{SphereFace2}}$ is the supervised speaker-classification loss from its SphereFace2 head; the pointwise and relational loss weights are 2 and 1. The teacher is frozen during student recovery; only the student, classification head, and projection are optimized.

4.5 Restart and weight interpolation

The student runs for 10,000 recovery steps. Both submitted extractors are linear interpolations of

two checkpoints with the same low-rank structure, so each is a single network of the original student size.

System A restarts training from the step-8,000 checkpoint for 5,000 additional steps with a new classification head, optimizer, learning-rate schedule, and random seed; the extractor weights carry over from recovery. The restart checkpoint is interpolated with its starting checkpoint:

$$\theta_A = 0.7\theta_{\text{restart},5000} + 0.3\theta_{\text{recovery},8000}. \quad (6)$$

System B interpolates the recovery step-6,000 checkpoint, selected by plain cosine EER (Section 6.1), with the compressed initialization:

$$\theta_B = 0.7\theta_{\text{recovery},6000} + 0.3\theta_{\text{init}}. \quad (7)$$

Here $\theta_{\text{restart},5000}$ denotes the parameters after 5,000 restart steps, $\theta_{\text{recovery},k}$ the parameters at recovery step k, and θ_{init} the compressed student before recovery. Unlike score-level fusion of two networks, interpolation yields a single network at inference.

5 Trial Scoring

System B scores a trial by the raw cosine score $s = \hat{e}_A^\top \hat{e}_B$ of whole-clip embeddings (Section 3), with no back-end. System A applies the following back-end to its extractor. Neither system uses cohort score normalization or test-set statistics: each trial score depends only on its two recordings.

For System A, each recording's embedding is the average of the length-normalized whole-clip and speech-only embeddings. A logistic scorer receives the pair's cosine score and quality features. For each of speech duration, SNR, and raw embedding norm, it computes the minimum, maximum, and product across the two sides. Calibration trials are built from five utterances per training speaker, using clean and noisy copies and excluding pairs from the same source file; no test recordings are used to fit it. The back-end feature recipe is designed with the teacher and retained for the student. Its learned logistic mapping converts the score and quality inputs to one scalar per trial. At inference, the back-end computes the quality features of both recordings and applies the fitted logistic mapping.

System	EER	minDCF
`E8s5000_wise0.7`, plain cosine	4.853	0.257
System A: `E8s5000_wise0.7` + back-end	4.449	0.271
System B: `E5s6000_wise0.7`, plain cosine	4.685	0.267

Table 2: Submitted systems on Public Test. EERs are percentages; minDCF uses target prior $p = 0.01$. The System A back-end is fitted on training speakers only; neither system uses score normalization.

Checkpoint	EER	minDCF
`E5s6000_wise0.7`	4.685	0.267
`E5s6000`	4.697	0.254
`E5s4000`	4.705	0.259
`E5s10000_wise0.7`	4.725	0.261
`E5s8000_wise0.7`	4.729	0.265
`E5s8000`	4.805	0.253
`E5s10000`	4.805	0.251
`E8s5000_wise0.7`	4.853	0.257
`E8s5000`	4.881	0.259

Table 3: Plain cosine scoring on Public Test (no back-end). E5sN: recovery step N; E5sN_wise0.7: $0.7\theta_{\text{recovery},N} + 0.3\theta_{\text{init}}$; E8s5000: restart step 5,000; E8s5000_wise0.7: θ_A (Section 4). Sorted by EER.

6 Results

6.1 Checkpoint selection

Table 3 compares nine candidate extractors by plain cosine scoring of whole-clip embeddings on Public Test, without any back-end. Candidates are recovery checkpoints at steps 4,000–10,000, their interpolations with the compressed initialization ($0.7\theta_{\text{recovery}} + 0.3\theta_{\text{init}}$), and the restart checkpoint with and without interpolation with its starting checkpoint. All candidates lie within 0.2 EER points of each other. By plain cosine, the interpolated recovery step-6,000 checkpoint is best (4.685%) and is submitted as System B. The restart checkpoints rank last under plain cosine, although they have competitive minDCF.

6.2 Submitted systems

Table 2 reports both submitted systems. System B is the plain-cosine winner of Table 3. System A uses `E8s5000_wise0.7`, whose plain-cosine EER is higher, but the back-end of Section 5 lowers its EER from 4.853% to 4.449%; minDCF rises slightly, from 0.257 to 0.271. System A therefore has the lower EER, while System B has the lower minDCF and needs no back-end.

7 Implementation and Deployment

At deployment, each system's extractor is one compressed network; System B adds only a cosine score. Teachers, classification heads, and KD projections are discarded, and the interpolated weights are materialized before inference. The extractor count in Table 1 excludes these training components. We have not measured edge CPU/NPU latency, peak activation memory, or energy.

Reproduction requires the cleaned training list, compression ranks, calibration inputs, crop bank and cached teacher targets, augmentation sources, and schedules, together with the recovery and restart weights, the compressed initialization, and interpolation coefficient 0.7. Each submission produces 278,410 finite trial scores in the supplied order. Training is not bitwise deterministic; exact scoring reproduction requires the submitted artifacts.

8 Conclusion

We presented a compact speaker verification system built by adapting and compressing ReDimNet2-B6. It combines short/long crop training, activation-whitened compression, pointwise and relational distillation, a training restart, weight interpolation, and quality-aware calibration. Each submitted extractor contains 6,947,208 parameters. With a back-end fitted on training speakers only, System A obtains 4.449% EER on Public Test; with plain cosine scoring, System B obtains 4.685%. Neither system uses score normalization. Only a single compressed embedding network is used at inference; the larger teacher is used only during training.

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